

Continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v)}{\partial x} = 0$$

$$G = \rho v A$$

$$\rho = \frac{p}{ZRT}$$

$$c^2 = ZRT$$

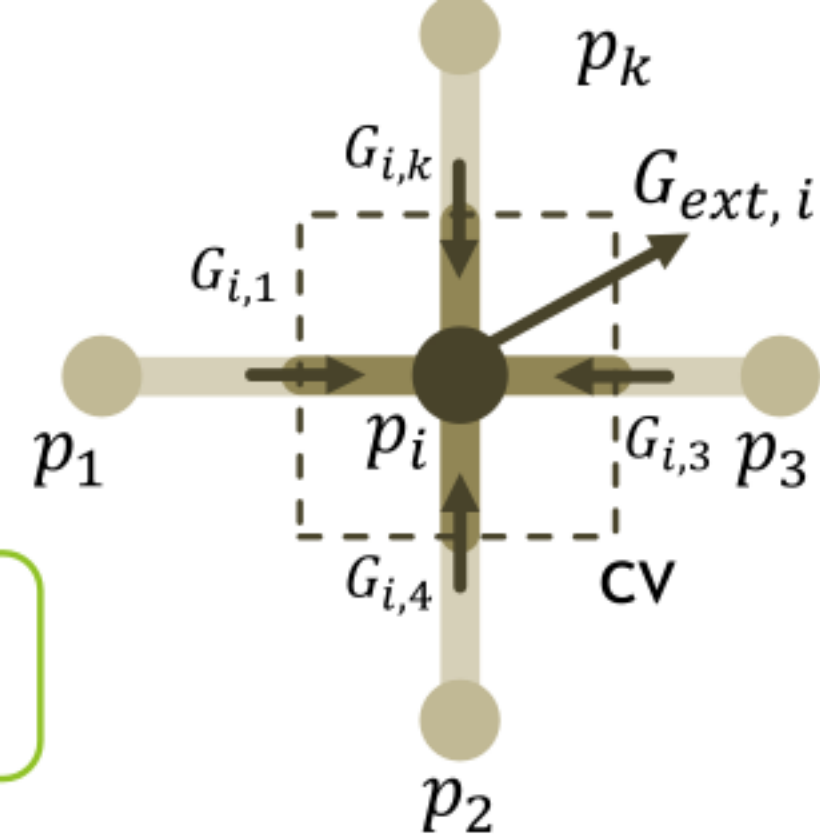
Control Volume integration

$$\frac{\partial p}{\partial t} = -\frac{c^2}{A} \frac{\partial G}{\partial x}$$

time discretization

$$\frac{V_i}{c_{i,j}^2} \frac{dp_i}{dt} = \sum_{j=1}^k G_{i,j} - G_{ext,i}$$

$$\frac{V_i}{c_{i,j}^2 \Delta t} (p_i^{n+1} - p_i^n) = \sum_{j=1}^k G_{i,j}^{n+1} - G_{ext,i}^{n+1}$$



Flow equation

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho v^2)}{\partial x} + \frac{\partial p}{\partial x} + \frac{\lambda}{2D} \rho v |v| + \rho g \sin \alpha = 0$$

creeping motion

horizontal pipelines

$$\frac{\partial p}{\partial x} = -\frac{\lambda c^2}{2DA^2 p} |G|G - \frac{1}{A} \frac{\partial G}{\partial t}$$

$$P = p^2$$

$$\Delta P_j^{n+1} = R_{fj} |G_j^{n+1}| G_j^{n+1} + R_{Ij} (G_j^{n+1} - G_j^n)$$

LINEARIZATION NEEDED!

Finite Difference over a pipeline

time discretization

with: Fluid-dynamic Resistance

$$R_f = \frac{16 \lambda c^2}{\pi^2 D^5 p} \Delta x$$

Inertia Resistance

$$R_I = \frac{2 p_m \Delta x}{A \Delta t}$$

Averaged over the pipeline length

- Friction factor λ
- Compressibility Factor $Z \rightarrow$ GERG-2008 wide range equation of state